

Concerns That the K4 Breakthrough Might Not Be Valid

In June, 2013 I announced to the Kryptos group the details of a breakthrough I made in K4 of Kryptos. I claimed a roughly 1:100,000,000 chance of the K4 breakthrough being random.

The response from the list was extremely helpful; a lot of concerns were brought up about the validity of the breakthrough. Obviously, if it is not really a breakthrough, it is in everyone's best interest to know.

So I looked carefully at each of the ~40 responses within the first 48 hours of my posting the details, looking for every possible position challenging the validity of my breakthrough. I broke them down into 5 different concerns:

- [1] Jim Sanborn said that no transposition was allowed (“NYPVTT->BERLIN 1:1” theory #1),
- [2] NYPVTT should not map to BERLIN in intermediate ciphertext (the other “NYPVTT->BERLIN 1:1” theory).
- [3] It's just alphabet soup, aside from BERLIN being in the right place,
- [4] There are a massive number of ways to get BERLIN in the right place, and
- [5] K4 should be solved using just 1 step.

I believe all these concerns (except the last one) are valid, but I personally do not believe any of them stand in the way of the K4 breakthrough being valid. I will address each one separately.

Concern #1: Jim Sanborn said that no transposition was allowed ("NYPVTT->BERLIN 1:1" theory #1)

Summary:

In October, 2011, there was a dinner with Jim Sanborn and a number of guests from the Kryptos group. The day after the dinner, someone posted that:

We asked about the last clue that had been released, and whether it was a one-to-one mapping from NYPVTT to BERLIN, or whether it was intended that this clue be a mapping of a six letter group to a six letter group. Mr. Sanborn indicated this is a one-to-one mapping.

This has been interpreted in several ways. One of the ways is that something is done with the letter "N" to turn it into "B" and that it always stays at position 64. If this is the case, it is said that transposition cannot be used in K4 (in which case the "N" would move somewhere else, and a different character would decode to "B").

My Thoughts:

The first thing to notice here is that Jim Sanborn has never publicly stated or implied this.

The second thing to notice is that there is much confusion on the Kryptos group as to what was meant at that dinner (there is another interpretation suggesting that Jim Sanborn meant simply that "BERLIN" appears in the plaintext where "NYPVTT" appears now).

The third thing to notice is that when the BERLIN clue was released, the author of the article implied that transposition was used ("the other 91 characters **and their proper order** are yet to be determined.").

The fourth is that if a group of people trying to solve Kryptos are confused as to what was said and meant at that dinner, Jim Sanborn could have easily misinterpreted the question.

If this concern is true, that transposition cannot be used, it would indeed mean that my K4 breakthrough is not real.

However, given all the confusion over a private conversation, I personally do not believe that transposition cannot be used in K4.

Concern #2: NYPVTT should not map to BERLIN in intermediate ciphertext (“NYPVTT->BERLIN 1:1” theory #2).

Summary:

See Concern #1 for details on the discussion from the October, 2011 dinner with Jim Sanborn.

One of the interpretations was that BERLIN appears in the plaintext, not the ciphertext of an intermediate decoding step. In other words, BERLIN is a word that Jim Sanborn used in whatever he wrote for K4, not a random group of letters that happened during decoding that happened to form the name of a city.

My Thoughts:

I do strongly believe that BERLIN appears in the plaintext. The New York Times article *implies* that that is the case, but does not state so directly. Also, this concern conflicts with Concern #5 (that there is only 1 step to K4). Both cannot be right.

More importantly, having BERLIN in the intermediate ciphertext does not mean that it does not appear in the plaintext. My belief is that a mask is used that causes some letters to be decoded, while others do not. This would allow BERLIN to appear in the intermediate decoding step as well as appearing in the plaintext. In fact, Ed Scheidt stated that a mask was used in K4.

So I do not really see a concern here, but in fact a major clue (that a mask is used that allows some characters to be decoded while others are not).

**Concern #3: It's just alphabet soup, aside from BERLIN being
in the right place.**

Summary:

This concern was brought up by several different people. There are two close varieties: [1] You shouldn't see alphabet soup (random characters) at all, you should see the final plaintext, and [2] If you do see random characters, BERLIN should not appear within them.

The assumption here is that either there is a single step to decoding K4, or that if multiple steps are used, they will all be just random characters until the entire plaintext is suddenly revealed.

My Thoughts:

First, if K4 involves multiple steps, then you are nearly guaranteed to see random characters in one or more intermediate steps. That quickly takes care of the first variety of the concern.

The second variety of the concern ties in with Concern #2. If K4 uses multiple steps and one of the steps is a mask that allows some characters to be decoded while others are not, it allows BERLIN to appear in the intermediate ciphertext as well as the plaintext. So that takes care of the other variety of this concern.

Also worth noting is the K3 plaintext, which ends with "Do you see anything Q?", which could be interpreted to mean that there is random text with one or more words visible inside.

**Concern #4: There are a massive number of ways to get BERLIN
in the right place.**

Summary:

Although we all know there are many ways of decoding K4 that get BERLIN in the correct location, one person suggested there was a massive number of ways to do so (suggesting billions of ways).

He did not elaborate, but suggested something along the lines of complex transposition (similar to one time pads), such as “Take the character from position 5 and put it in position 64, then take the character from position 19 and put it in position 65, ...”

My Thoughts:

There could possibly be an infinite number of ways of getting BERLIN in the correct location, using creative transposition methods and multiple steps.

But, this does not in any way suggest that the K4 breakthrough is invalid. It just shows that the more creative you are allowed to get (the more steps you take, the more you diverge from standard cryptography, etc.), the more likely you are to find ways to force BERLIN in the right place.

Since I'm not talking about something far-fetched, this doesn't apply (and even if it *did*, it doesn't suggest that the K4 breakthrough is invalid, just that it is to some degree less likely than the ~1:100,000,000 odds I came up with). The K4 breakthrough is very straightforward; combining the methods used in K1-K3, and using key words and key numbers that appear in what Jim Sanborn created.

Finally, according to this concern, essentially *no* multi-step solution would be worth pursuing. So this concern really cannot be used as the basis of not pursuing the K4 breakthrough.

Concern #5: K4 should be solved using just 1 step.

Summary:

At least 2 people suggested that K4 should be solved in just one step, and that there should be no intermediate ciphertext.

My Thoughts:

I find this line of thought very surprising (and limiting).

First, it should be noted that this concern conflicts with Concern #2 – both cannot be true.

As far as I know, Jim Sanborn has not said anything suggesting that this is the case. And there are statements suggesting otherwise (e.g. the New York Times article mentioning the BERLIN clue stated “There are many steps to cracking the code”, Jim Sanborn stated that he thinks he used 5-6 cryptographic techniques in Kryptos, Jim Sanborn discussing in the NOVA special how multiple steps would let the code get very complex very quickly). Even the NSA in 1992 suggested the possibility of multiple steps.

So if someone believes, for whatever reason, that K4 is solved in just one step, they have that belief, and there isn't really a way to counter it. They will simply choose not to pursue multi-step solutions.

Since nothing has been said stating that there should be just one step, however, I see no reason to assume that it must be true.